



Exploratory data analysis of Book sales

(Kaggle [Dataset](#))

Python analysis by Elodie Barthout
Last update: July 10th, 2026

Content

Dataset description	03
Data cleaning	04
General analysis	05
Exploratory analysis	09
Focus on Stephen King	11
Focus on John Grisham	15
Focus on Nicholas Sparks	19
Focus on Janet Evanovich	34
Focus on Lemony Snicket & Brett Helquist	28
Conclusion	31

Dataset description

This dataset provides comprehensive information on various aspects of books, including:

- Publishing year
- Language of publication
- Author details
- Ratings given by readers
- Sales performance data
- Genre classification

Data cleaning

Before cleaning:

- 1070 entries
- 4 genres
- Publication year ranging from 560 BC to 2016

Cleaning process:

- Removing null & unreadable values
- Fixing inconsistencies
- Checking duplicate records
- Filtering publication year

After cleaning:

- 1015 entries
- 3 genres
- Publishing year ranging from 1812 to 2016

General analysis

Modern books published between 1812 & 2016

- > 99% of them published in **english**
- > The vast majority being **fiction** books
- > All genres being rated the same on average (4,0/5)

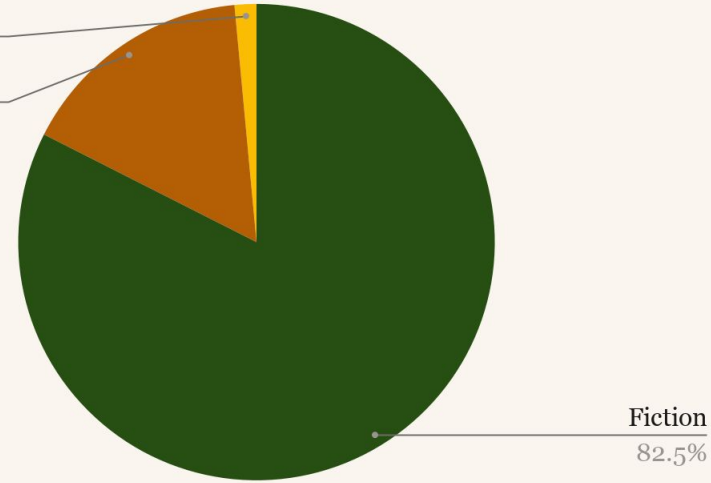
Genre distribution

Children

1.5%

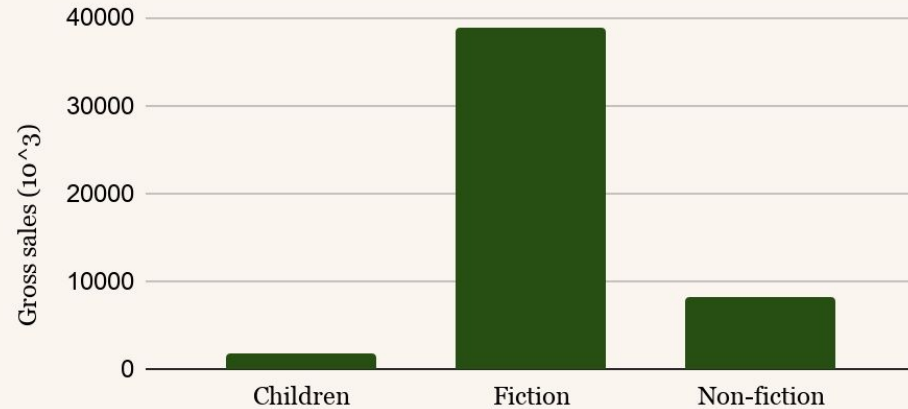
Non-fiction

16.1%

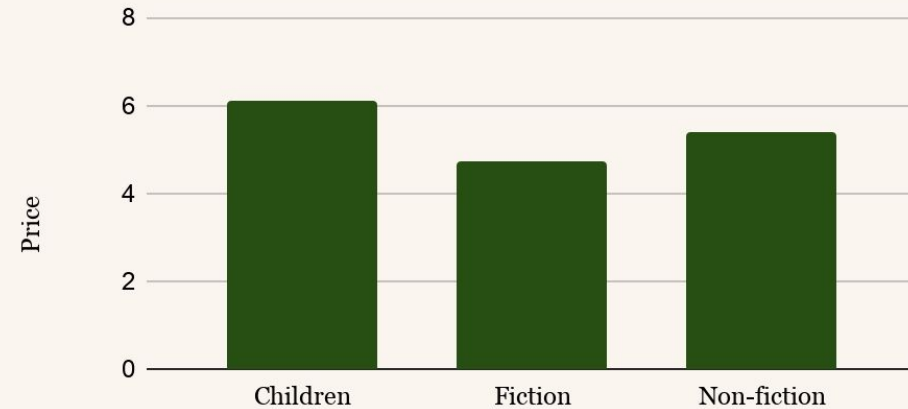


General analysis

Which genre generated the most revenue?



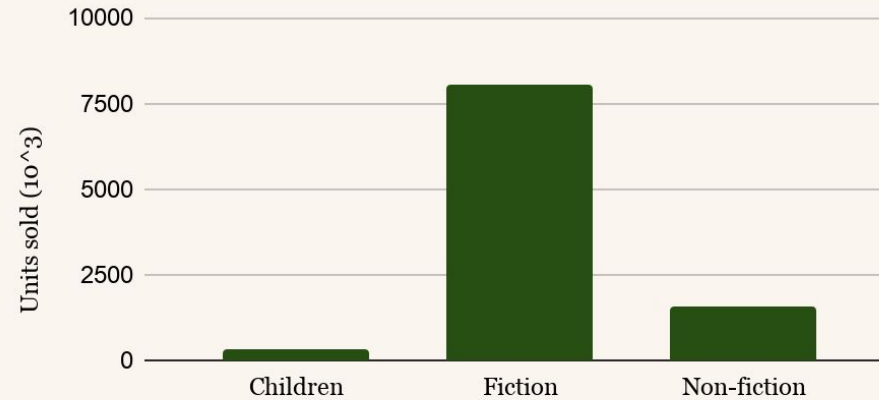
Average book price depending on genre



Fiction is the genre which generated the most revenue despite being the **cheapest** on average.

General analysis

Which genre is the most purchased?



This is because fiction is the **best-selling genre**, with sales volume more than 5 times higher than other genres.

General analysis

What was the **best-selling book** in each genre?

Genre	Title	Author	Year of publication	Units sold
Children	Crytonomicon	Neal Stephenson	1999	50 112
Fiction	New Moon: The Complete Illustrated Movie Companion	Mark Cotta Vaz	2009	61 560
Non-fiction	A Crown of Swords	Robert Jordan	1996	54 000

NB: New Moon: The Complete Illustrated Movie Companion being the best-selling book of this dataset.

Exploratory analysis

Top 3 authors with the **most books published**:

1. Stephen King (29 books)
2. John Grisham (13 books)
3. Nicholas Sparks (10 books)

Exploratory analysis

Top 3 authors with the most books published:

1. **Stephen King (29 books)**
2. John Grisham (13 books)
3. Nicholas Sparks (10 books)

Focus on Stephen King

- 29 books published between **1979 & 2015**
- Rated mostly as an intermediate writer during that time
- Multiple books published in 1980, 1987, 1999, 2013 & 2014
- Mostly wrote **fiction** books (24/29 books)
- Sold on average 9 597 books
- Books rated **3,89/5** on average
- Average sale price of \$3,68

Focus on Stephen King

Top 3 **best-selling** books:

- Skeleton Crew in 1985 (371 761 units sold)
- The Tommyknockers in 1987 (304 052 units sold)
- Joyland in 2013 (42 340 units sold)

> All fiction books.

Focus on Stephen King

Top-rated book:

Different seasons published in 1982

- > fiction book rated 4,34/5
- > 5000 units sold

Lowest-rated book:

Gerald's Game published in 1992

- > fiction book rated 3,47/5
- > 5886 units sold

Exploratory analysis

Top 3 authors with the most books published:

1. Stephen King (29 books)
2. **John Grisham (13 books)**
3. Nicholas Sparks (10 books)

Focus on John Grisham

- 13 books published between **1994 & 2014**
- Rated as an intermediate writer during that time
- Mostly wrote **fiction** books (8/13 books)
- Sold on average 7 048 books
- Books rated **3,78/5** on average
- Average sale price of \$4,3

Focus on John Grisham

Top 3 **best-selling books**:

- The Brethren published in 2000 (34 992 units sold)
- The Broker published in 2005 (29 592 units sold)
- The Partner published in 1997 (5 049 units sold)

> All fiction books.

Focus on John Grisham

Top-rated book:

Sycamore Row published in 2013

- > fiction book rated 3,98/5
- > 3 969 units sold

Lowest-rated book:

Gray Mountain published in 2014

- > non-fiction book rated 3,52/5
- > only 106 units sold

Exploratory analysis

Top 3 authors with the most books published:

1. Stephen King (29 books)
2. John Grisham (13 books)
3. **Nicholas Sparks (10 books)**

Focus on Nicholas Sparks

- 10 books published between **2000 & 2012**
- Rated mostly as an intermediate writer during that time
- Published multiple books in 2003
- Only wrote **fiction** books
- Sold on average 2 073 books
- Books rated **3,97/5** on average
- Average sale price of \$2,29

Focus on Nicholas Sparks

Top 3 **best-selling** books:

- True Believer in 2003 (4 440 units sold)
- At First Sight in 2004 (4 320 units sold)
- The Longest Ride in 2012 (3 942 units sold)

> All fiction books.

Focus on Nicholas Sparks

Top-rated book:

The Guardian published in 2003

- > fiction book rated 4,14/5
- > 1 892 units sold

Lowest-rated book (which is also their best-selling one):

True Believer published in 2003

- > fiction book rated 3,8/5
- > 4 440 units sold

- > Both books have been published the **same year**.

Exploratory analysis

Top 3 **best-selling authors**:

1. Stephen King (278 322 books sold)
2. Janet Evanovich (181 089 books sold)
3. Lemony Snicket & Brett Helquist (158 973 books sold)

Exploratory analysis

Top 3 best-selling authors:

1. Stephen King (278 322 books sold)
2. **Janet Evanovich (181 089 books sold)**
3. Lemony Snicket & Brett Helquist (158 973 books sold)

Focus on Janet Evanovich

- 9 books published between **1996 & 2005**
- Rated as an excellent writer during that time
- Mostly wrote **fiction** books (7/9 books) but a non-fiction book as well as a children book
- Sold on average 20 121 books
- Books rated **4,17/5** on average
- Average sale price of \$6,58

Focus on Janet Evanovich

Top 3 **best-selling books**:

- Hot Six in 2000 (48 600 units sold)
- High Five in 1999 (43 200 units sold)
- Seven Up in 2001 (36 072 units sold)

> All fiction books except High Five, which is a children book.

Focus on Janet Evanovich

Top-rated book:

Janet Evanovich Three and Four Two-Book Set published in 2002

- > fiction book rated 4,34/5
- > 2 889 units sold

Lowest-rated book (being her only non-fiction book):

Two for the Dough published in 1996

- > non-fiction book rated 4,11/5
- > 5 886 units sold

Summary

1. Stephen King

29 books published in 36 years (0,8 book/year) with, on average:
9 500 units sold & a rating of 3,89/5

2. John Grisham

13 books published in 20 years (0,65 book/ year) with, on average:
7 000 units sold & a rating of 3,78/5

3. Nicholas Sparks

10 books published in 12 years (0,83 book/year) with, on average:
2 000 units sold & a rating of 3,97/5

Exploratory analysis

Top 3 best-selling authors:

1. Stephen King (278 322 books sold)
2. Janet Evanovich (181 089 books sold)
3. **Lemony Snicket & Brett Helquist (158 973 books sold)**

Focus on co-authors Lemony Snicket & Brett Helquist

- 8 books published between **1999 & 2003**
- Rated mostly as intermediate writers during that time
- Mostly wrote **fiction** books (7/8 books)
- Published multiple books in 2000 & 2001
- Sold on average 19 871 books
- Books rated **3,95/5** on average
- Average sale price of \$4,49

Focus on co-authors Lemony Snicket & Brett Helquist

Top 3 **best-selling** books:

- The Miserable Mill in 2000 (46 872 units sold)
- The Austere Academy in 2000 (36 288 units sold)
- The Vile Village in 2001 (35 640 units sold)

> Two of them have been published the **same year**.

Focus on co-authors Lemony Snicket & Brett Helquist

Top-rated book (*being their only non-fiction book*):

The Ersatz Elevator published in 2001

- > non-fiction book rated 4,01/5
- > 31 104 units sold

Lowest-rated book (*being also their best-seller*):

The Miserable Mill published in 2000

- > fiction book rated 3,83/5
- > 46 872 units sold

Summary

1. Stephen King

29 books published in 36 years (0,8 book/year) with, on average:
9 500 units sold & a rating of 3,89/5

2. Janet Evanovich

9 books published in 9 years (1 book/ year) with, on average:
20 000 units sold & a rating of 4,17/5

3. Lemony Snicket & Brett Helquist

8 books published in 4 years (2 books/year) with, on average:
19 800 units sold & a rating of 3,95/5

Conclusion

- All genres have been rated **4,0/5** on average meaning that they are equally appreciated among readers.
- **Fiction** is the genre which generated the most revenue despite being the cheapest. This is because fiction is also the best-selling genre with at least 5 times more sales than other genres. The best-selling book of this dataset is a fiction book as well.
 - > This genre represents **82,5% of this dataset** which probably explains those numbers.
- **Stephen king** is the best-selling author but also the most prolific one with the most books published (29) & the largest publication window (36 years), which might explain its best-selling position.